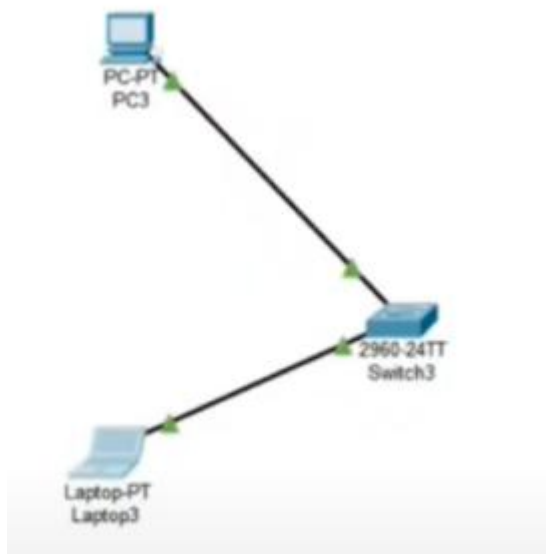


Computer Networks: Assignment-2

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Q 1) Construct a virtual LAN in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.



IP config:

PC3: 192.168.10.2 / 255.255.255.0

Laptop3: 192.168.10.3 / 255.255.255.0

Output:

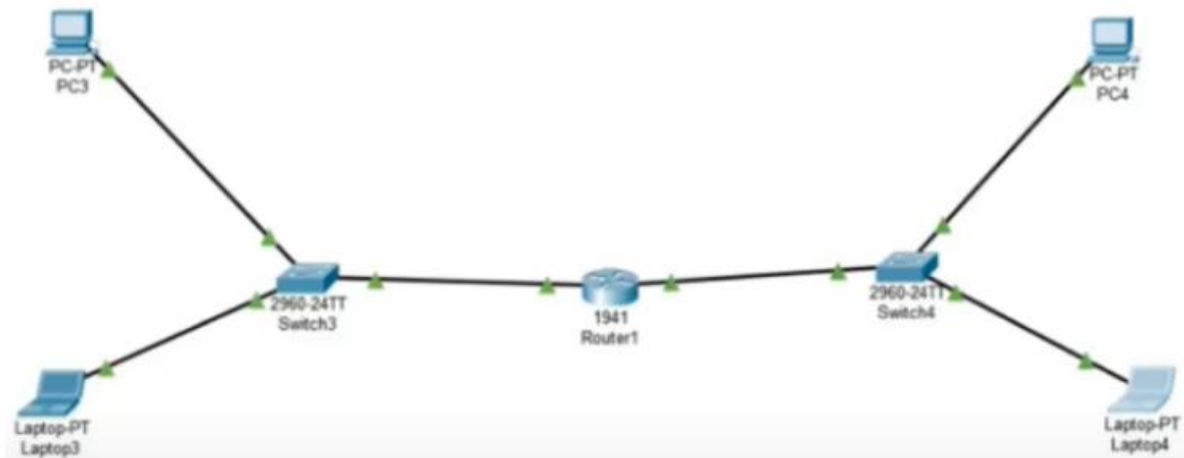
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms
```

Q 2) Construct a virtual MAN in the Cisco packet tracer, as shown below, and check where the data packets transfer from one system to another.



IP Config:

PC3: 192.168.1.2, GW 192.168.1.1

Laptop3: 192.168.1.3, GW 192.168.1.1

PC4: 192.168.2.2, GW 192.168.2.1

Laptop4: 192.168.2.3, GW 192.168.2.1

Commands:

For router:

```
enable
configure terminal

interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

end
write memory
```

Output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

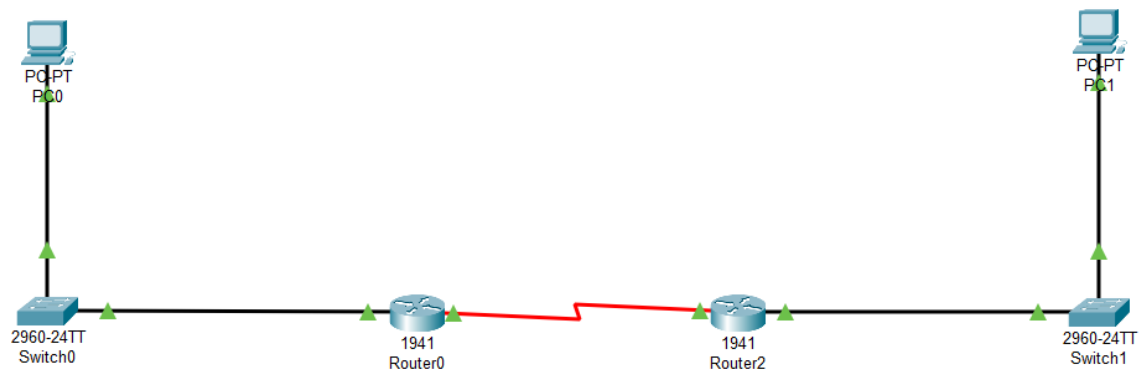
Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Q 3) Construct a virtual WAN in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

Diagram:



IP Config:

PC0: 192.168.10.2 / 255.255.255.0, GW 192.168.10.1

PC1: 192.168.20.2 / 255.255.255.0, GW 192.168.20.1

Commands:

For Router1:

```
enable
configure terminal

interface gigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface serial0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit

ip route 192.168.20.0 255.255.255.0 10.0.0.2
end
write memory
```

For Router2:

```
enable
```

configure terminal

interface gigabitEthernet0/0

ip address 192.168.20.1 255.255.255.0

no shutdown

exit

interface serial0/0/0

ip address 10.0.0.2 255.255.255.252

no shutdown

exit

ip route 192.168.10.0 255.255.255.0 10.0.0.1

end

write memory

Output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

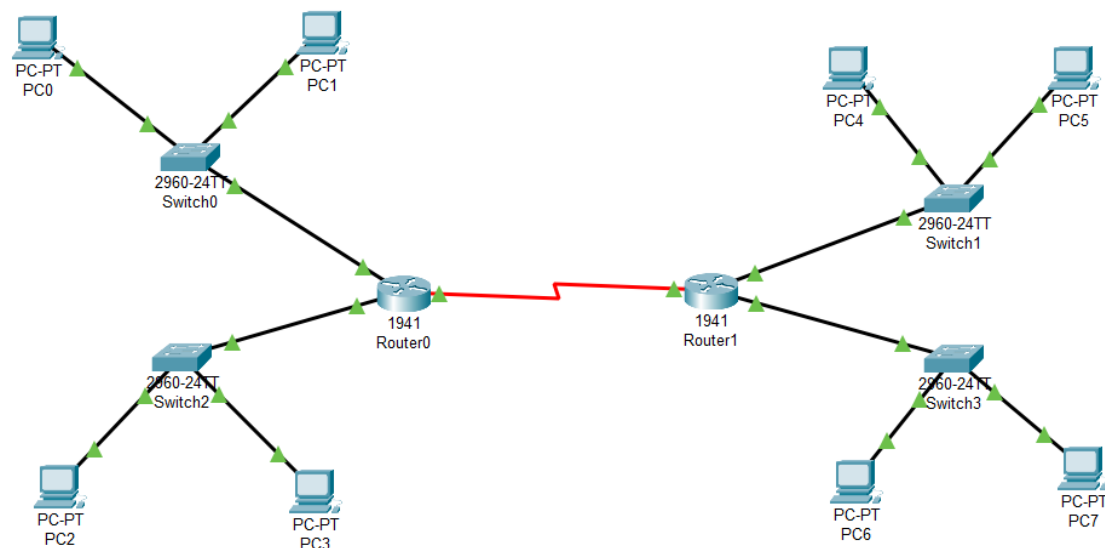
Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=6ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 6ms
```

Q-4 In Cisco Packet Tracer, design a campus network with two buildings (Building-A and Building-B). Each building should have at least two IP subnets (e.g., Subnet 10–Students and Subnet 20–Faculty) implemented using switches. Connect both buildings via routers (using either subinterfaces or inter-router links), assign suitable IP subnets to each group, configure links between switches and routers, set up inter-subnet routing so that only Faculty subnets of both buildings can communicate with each other while Student subnets remain isolated, and verify this behavior using ping and simulation mode to observe packet flow

Diagram:



IP Config:

A-Student-PC0: 192.168.10.10 / 255.255.255.0, GW 192.168.10.1

A-Student-PC1: 192.168.10.11 / 255.255.255.0, GW 192.168.10.1

A-Faculty-PC2: 192.168.20.10 / 255.255.255.0, GW 192.168.20.1

A-Faculty-PC3: 192.168.20.11 / 255.255.255.0, GW 192.168.20.1

B-Student-PC4: 192.168.30.10 / 255.255.255.0, GW 192.168.30.1

B-Student-PC5: 192.168.30.11 / 255.255.255.0, GW 192.168.30.1

B-Faculty-PC6: 192.168.40.10 / 255.255.255.0, GW 192.168.40.1

B-Faculty-PC7: 192.168.40.11 / 255.255.255.0, GW 192.168.40.1

Commands:

For Router1:

```
interface GigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

```
interface GigabitEthernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown

interface Serial0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000 !
no shutdown

ip route 192.168.40.0 255.255.255.0 10.0.0.2
```

For Router2:

```
interface GigabitEthernet0/0
ip address 192.168.30.1 255.255.255.0
no shutdown

interface GigabitEthernet0/1
ip address 192.168.40.1 255.255.255.0
no shutdown

interface Serial0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown

ip route 192.168.20.0 255.255.255.0 10.0.0.1
```

Output:

 PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

 PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.40.10: bytes=32 time<1ms TTL=126
Reply from 192.168.40.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```